

SANSKAR TIWARI

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B.Tech ENTC @ PICT (CGPA: 9.30) · 3× National/International Hackathon Winner · 2.15L+ in prizes
Building production AI systems at the intersection of Computer Vision, LLM Orchestration, and Embedded Edge Intelligence.

Education

Pune Institute of Computer Technology (PICT) <i>B.Tech, Electronics and Telecommunication Engineering — CGPA: 9.30</i> — Relevant Coursework: Data Structures & Algorithms, Digital Electronics, Signals & Systems, Microprocessors, Probability, Linear Algebra.	Jul 2024 – Present <i>Pune, India</i>
Arya Gurukul International College of Science and Commerce <i>HSC: 82.33% MHTCET: 99.34 Percentile JEE Mains: 97.03 Percentile</i>	Jul 2022 – Apr 2024 <i>Navi Mumbai, India</i>

Achievements

TechFiesta '26 International Hackathon — 1st Rank, Agriculture Domain <i>700+ Teams, 3500+ Participants · Grand Prize 50,000</i> — Won 1st place among 700+ national and international teams for KrishiSahAI — an AI-powered agricultural intelligence platform combining computer vision, generative advisory, and real-time contextual data. — Mid-hackathon, scrapped and completely redesigned backend architecture within 10 hours after sponsor challenge; rebuilt inference pipeline and farmer-facing UI from scratch under pressure — final system passed industry-level scrutiny. — Led technical architecture, AI model integration, and final pitch to expert jury; prototype selected for real-world deployment evolution into KrishiSahAI.	Mar 2026 <i>Pune, India</i>
VOIS Innovation Marathon 2.0 (Vodafone Idea Foundation) — Top 3 National Finalist <i>630+ Teams · Team Prize: 2,00,000</i> — Selected Top 3 nationally among 630+ teams over a 5-month competitive cycle (Aug 2025 – Feb 2026) for KrishiSaarthi, an AI + blockchain agricultural advisory platform. — Conducted field research across 15+ agri-businesses in Baramati and Pune to validate the problem before building — chose farmer-first UX as core design constraint throughout development. — Engineered modular backend integrating deep learning crop advisory, smart contract-based green credit workflow, and LangChain-powered business strategy engine.	Mar 2026 <i>Pune, India</i>
Pune Agri International Hackathon — National Runner-Up <i>Competing against established startups and companies · 15L government development grant</i> — Secured Runner-Up against professional startups and companies as one of the very few student teams; presented before Maharashtra CM Devendra Fadnavis, Agriculture Minister, and senior government officials. — Built Krishi Prabandh — AI-powered agricultural governance layer with geo-tagged video claim verification, OCR fraud detection, satellite NDVI validation, and real-time officer dashboards.	May 2026 <i>Pune, India</i>

Experience

MindstriX Technologies LLP <i>Software Engineer Intern</i> — Building and deploying AI-powered backend services and LLM integration pipelines within the Technology and Product Development team. — Contributing to API development for intelligent applications, gaining hands-on experience shipping production-grade AI features end-to-end. — Collaborating on R&D initiatives in emerging AI technologies under experienced engineers.	Mar 2026 – Present <i>Nanded, Maharashtra (Remote)</i>
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Projects

KrishiSahAI Advisory — 📺 Demo Successor of KrishiSaarthi — Architecting full-stack AI agriculture platform with disease classification (38 classes, CNN), real-time pest detection (YOLOv8), multilingual voice interaction, weather intelligence, and LLM-powered business advisory. — Designed and implemented ROI-based multi-year business strategy engine using LangChain + LLM orchestration; generates contextual advisory from live WeatherAPI and agri-news feeds. — Built modular Flask REST backend with Firebase authentication; designed for production scalability from the ground up after mid-hackathon architecture pivot under 10-hour deadline.	Aug 2025 – Ongoing
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Intelligent Pothole Detection System —  GitHub

Jan 2026 – Mar 2026

- Identified I²C blocking conflicts in single-node design; redesigned into distributed dual-node architecture (ESP32-CAM for vision, ESP32 for sensors) — eliminated concurrency issues and stabilized real-time streaming.
- Trained YOLOv8-Nano pothole detector (mAP: **81.68%**, Precision: **82.33%**); integrated SORT tracking for frame-to-frame consistency and duplicate suppression.
- Built event-driven sensor fusion pipeline: MPU6050 accelerometer + NEO-6M GPS + DS3231 RTC triggered at detection events to compute severity scores and log geo-stamped structured outputs.
- 50+ hours of iterative prototyping addressing network latency, sensor noise, and false positives from shadows and road markings; IEEE research paper in preparation.

Dynamic Traffic Signal System —  GitHub

Jul 2025 – Nov 2025

- Built adaptive traffic control system using YOLOv8 + OpenCV; chose density-based dynamic timing (30–90s green window) over fixed-cycle logic, demonstrably reducing per-lane queue length under simulated load.
- Implemented 5-second sliding window vehicle density algorithm with JSON-based configuration for flexible parameter tuning; integrated ESP32 hardware control layer via TCP/IP.
- Demonstrated stable Python-to-ESP32 communication with reliable real-time detection as a smart-city proof-of-concept for intelligent traffic management.

Sunlight Tracking Mechanism —  GitHub

Feb 2025 – Mar 2025

- Engineered solar tracking prototype using dual LDR sensors, analog comparison logic, and servo motor feedback loop for continuous angular alignment under varying light and shadow conditions.
- Tech Stack: Arduino, Embedded C, Servo Control, Sensor Integration.

Research

- **Intelligent Pothole Detection using Computer Vision, Multi-Object Tracking, and Sensor Fusion** — IEEE paper in preparation (2026). Evaluates distributed dual-node architecture, YOLOv8-Nano performance (mAP 81.68%), and real-time sensor fusion methodology.

Technical Skills

Core:	Python, C++, YOLOv8, LangChain, LlamaIndex, ChromaDB, FastAPI, Flask
AI/ML:	CNN, Object Detection, SORT Tracking, Sensor Fusion, RAG, LLM Orchestration, TensorFlow
Embedded:	ESP32, ESP32-CAM, MPU6050, NEO-6M GPS, DS3231 RTC, Arduino, Embedded C
Backend/DB:	REST APIs, Firebase, Node.js, SQL
Frontend:	React, TypeScript, JavaScript, HTML, CSS
Tools:	Git, Linux, Docker, VS Code, Arduino IDE
Exposure:	Kotlin, Android (Gemini API), Blockchain/Smart Contracts, OpenCV

Learning & Specializations

- **Machine Learning Specialization** — Stanford University & DeepLearning.AI (Andrew Ng, 2025); applied concepts in computer vision, predictive modeling, and AI system development.
- **Advanced AI Systems:** Hands-on learning in Retrieval-Augmented Generation (RAG), LangChain, LlamaIndex, ChromaDB, Vector Search, and LLM-powered application architecture.

Competitive Programming

- HackerRank: **5-Star in C++**, 4-Star in C. Solved **250+ problems** across LeetCode and HackerRank — graphs, dynamic programming, greedy, recursion.